

MATH 3210-004 PSet 7

Specification

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1 Background

This problem set continues to develop integral calculus. More specifically, the first problem focuses on approximating integrals using Riemann sums. The next three problems study finer properties of Riemann integrable functions and Riemann integrals, with an emphasis on integrals of continuous functions. Recall that any continuous function on a compact interval is Bourbaki integrable (after [problem set 5](#)) and any Bourbaki integrable function on a compact interval is Riemann integrable (after [problem set 6](#)), hence for continuous functions, the integrals in question can be thought of either Bourbaki integrals or Riemann integrals. Finally the last two problems transition to the concept of a gauged partition, a concept foundational for the more advanced [Henstock-Kurzweil integration](#).

Gauge Let I be a compact interval. By definition a **gauge**¹ is a positive function on I , and the collection of all gauges are denoted by

¹There are (at least) two other uses of the word "gauge" in mathematics and sciences, one of them in [functional analysis](#) and the other in [\(mathematical\) physics](#); these are notions that are not quite related to the way we use the word "gauge", at least not immediately.

$\mathcal{G}(I)$ (thus $\mathcal{G}(I) = F(I; \mathbb{R}_{>0})$). For any $\delta \in \mathcal{G}(I)$ and for any $(\mathcal{P}, s) \in \text{sPar}(I)$, $(\mathcal{P}, s)$ is δ -**fine** if for any $P \in \mathcal{P}$, P is contained in the closed ball $I[s(P) \mid \leq \delta \circ s(P)]$ centered at $s(P)$ of radius $\delta \circ s(P)$, that is,

$$\boxed{\forall P \in \mathcal{P}, \forall x \in P : |x - s(P)| \leq \delta \circ s(P)}.$$

A **gauged partition** of I by definition is a triple $(\mathcal{P}, s, \delta)$ such that

- $(\mathcal{P}, s)$ is a sampled partition of I ,
- δ is a gauge on I , and
- $(\mathcal{P}, s)$ is δ -fine.

For $(\mathcal{P}, s, \delta), (\mathcal{Q}, t, \gamma) \in \text{gPar}(I)$, we write

- $\boxed{(\mathcal{P}, s, \delta) \succ_{\text{mesh}}(\mathcal{Q}, t, \gamma)}$, if for any $x \in I$, $\text{mesh}(\mathcal{P}) \leq \text{mesh}(\mathcal{Q})$.
- $\boxed{(\mathcal{P}, s, \delta) \succ_{\text{ref}}(\mathcal{Q}, t, \gamma)}$, if for any $P \in \mathcal{P}$, there is a $Q \in \mathcal{Q}$ such that $P \subseteq Q$.
- $\boxed{(\mathcal{P}, s, \delta) \succ_{\text{gauge}}(\mathcal{Q}, t, \gamma)}$, if for any $x \in I$, $\delta(x) \leq \gamma(x)$.

2 Generative AI and Computer Algebra Systems Regulations

This section applies only if you choose to use either a **generative AI tool** (e.g. chatbots) or a **computer algebra system (CAS)** while working on this problem set. If you do not use such tools, you may skip this section.

The use of these tools while completing this problem set is permitted, provided it is done responsibly and in a manner that supports your learning. Note however that in exams you will not be allowed to use any AI or CAS.

2.1 Disclosure

If you use such tools, you must disclose this fact in the designated section of the form you will complete as part of your submission. You must also share your reflections on which parts of the problem required your own mathematical judgment, insight, or decision-making, and which parts could reasonably be delegated to a computational or generative tool.

2.2 Guidelines for Responsible Use

If you use a generative AI tool or chatbot, you must adhere to the following guidelines:

- During the chat you may share with the chatbot parts or all of this specification document, as well as parts of the textbook or other sources.
- Directly asking the tool for complete problem solutions is prohibited.
- Appropriate uses include asking for conceptual explanations, checking intermediate steps, clarifying definitions or theorems, or exploring alternative solution approaches, provided that the final work submitted is your own.
- Regardless of the tools used, you are expected to understand, justify, and be able to reproduce all submitted work independently. The use of AI or CAS does not reduce or replace this expectation.

3 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details.

When tackling statements in MATH 3210, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false".

Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example, you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. Such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let $I = [0, 1]$, $f : I \rightarrow \mathbb{R}$, $x \mapsto x^2$, and put $\varepsilon = 10^{-2} \in \mathbb{R}_{>0}$ and $R_* = 1/3 \in \mathbb{R}$.

(a) Compute the least $n \in \mathbb{Z}_{\geq 1}$ such that for the following sampled partition $(\mathcal{P}_n, s_n)$ of I , the associated Riemann sum $R(f; \mathcal{P}_n, s_n)$ is not more than ε distant from R_* :

$$\mathcal{P}_n = \left\{ \left[\frac{k}{n}, \frac{k+1}{n} \right] \mid k \in \{0, 1, \dots, n-1\} \right\},$$

$$s_n : \left[\frac{k}{n}, \frac{k+1}{n} \right] \mapsto \frac{k+1/2}{n}.$$

(b) Compute the least $n \in \mathbb{Z}_{\geq 1}$ such that for the following sampled partition $(\mathcal{Q}_n, t_n)$ of I , the associated Riemann

sum $R(f; \mathcal{Q}_n, t_n)$ is not more than ε distant from R_* :

$$\mathcal{Q}_n = \left\{ \left[\frac{k}{2^n}, \frac{k+1}{2^n} \right] \mid k \in \{0, 1, \dots, 2^n - 1\} \right\},$$

$$t_n : \left[\frac{k}{2^n}, \frac{k+1}{2^n} \right] \mapsto \frac{k}{2^n}.$$

(c) Compute a partition $\mathcal{R}$ of I such that there are two sampling functions $p, q : \mathcal{R} \rightarrow I$ such that

$$|R(f; \mathcal{R}, p) - R_*| \leq \varepsilon \leq |R(f; \mathcal{R}, q) - R_*|.$$

(d) Compute a partition $\mathcal{S}$ of I such that for any sampling function $r : \mathcal{S} \rightarrow I$,

$$|R(f; \mathcal{S}, r) - R_*| \leq \varepsilon.$$

2. Let $f \in C^0(\mathbb{R}; \mathbb{R})$ be strictly increasing and $f(0) = 0$. Then

(a) For any $\alpha, \beta \in \mathbb{R}_{>0}$, $\alpha\beta \leq \int_{[0, \alpha]} f(x)dx + \int_{[0, \beta]} f^{-1}(x)dx$.

(b) For any $\alpha, \beta \in \mathbb{R}_{>0}$, $\alpha\beta = \int_{[0, \alpha]} f(x)dx + \int_{[0, \beta]} f^{-1}(x)dx$ iff $\beta = f(\alpha)$.

3. Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$.

(a) For any $f \in R(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x)dx.$$

(b) For any $f \in C^0(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x)dx.$$

- (c) For any $f \in C^0(I; \mathbb{R})$ and for any $g \in R(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$\int_I f(x)g(x)dx = f(x_*) \int_I g(x)dx.$$

- (d) For any $f \in C^0(I; \mathbb{R})$ and for any $g \in R(I; \mathbb{R}_{\geq 0})$, there is an $x_* \in]\lambda, \rho[$ such that

$$\int_I f(x)g(x)dx = f(x_*) \int_I g(x)dx.$$

4. Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$, $f \in R(I; \mathbb{R})$. Then

- (a) There is a $\mu \in I$ such that $\int_{[\lambda, \mu]} f(x)dx = \int_{[\mu, \rho]} f(x)dx$.
 (b) There is a $\mu \in]\lambda, \rho[$ such that $\int_{[\lambda, \mu]} f(x)dx = \int_{[\mu, \rho]} f(x)dx$.

5. Let I be a compact interval. Then

- (a) For any $(\mathcal{P}, s) \in \text{sPar}(I)$, there is a $\delta \in \mathcal{G}(I)$ such that $(\mathcal{P}, s)$ is δ -fine.
 (b) For any $\delta \in \mathcal{G}(I)$, there is a $(\mathcal{P}, s) \in \text{sPar}(I)$ such that $(\mathcal{P}, s)$ is δ -fine.
 (c) For any finite set $S \subseteq I$, there is a $\delta \in \mathcal{G}(I)$ such that for any $(\mathcal{P}, s) \in \text{sPar}(I)$, if $(\mathcal{P}, s)$ is δ -fine, then $S \subseteq \text{im}(s)$.
 (d) For any finite set $S \subseteq I$, there is a $\delta \in \mathcal{G}(I)$ such that there is a δ -fine $(\mathcal{P}, s) \in \text{sPar}(I)$ with the property that $S = \text{im}(s)$.
 (e) $\succsim_{\text{mesh}}$, $\succsim_{\text{ref}}$ and $\succsim_{\text{gauge}}$ are all directed set structures on the collection $\text{gPar}(I)$ of all gauged partitions on I .
 (f) For any two distinct $\succsim_1, \succsim_2 \in \{\succsim_{\text{mesh}}, \succsim_{\text{ref}}, \succsim_{\text{gauge}}\}$, there are $(\mathcal{P}, s, \delta), (Q, t, \gamma) \in \text{gPar}(I)$ such that $(\mathcal{P}, s, \delta) \succsim_1 (Q, t, \gamma)$ but $(\mathcal{P}, s, \delta) \not\succsim_2 (Q, t, \gamma)$.

6. Let I be a compact interval, $(\mathcal{P}, s, \delta) \in \text{gPar}(I)$. Then

(a) there is a $(\mathcal{Q}, t) \in \text{sPar}(I)$ such that

- $(\mathcal{Q}, t)$ is δ -fine,
- $\forall x \in \text{im}(t), \exists ! Q \in \mathcal{Q} : x \in Q$ ²,
- for any $f \in F(I; \mathbb{R})$, $R(f; \mathcal{P}, s) = R(f; \mathcal{Q}, t)$.

(b) there is a $(\mathcal{Q}, t) \in \text{sPar}(I)$ such that

- $(\mathcal{Q}, t)$ is δ -fine,
- $\forall x \in \text{im}(t), \exists Q_1, Q_2 \in \mathcal{Q} : Q_1 \cap Q_2 = \{x\}$ ³,
- for any $f \in F(I; \mathbb{R})$, $R(f; \mathcal{P}, s) = R(f; \mathcal{Q}, t)$.

(c) there is a $(\mathcal{Q}, t) \in \text{sPar}(I)$ such that

- $(\mathcal{Q}, t)$ is δ -fine,
- $\forall Q \in \mathcal{Q} : t(Q) \in \{\inf(Q), \sup(Q)\}$ ⁴,
- for any $f \in F(I; \mathbb{R})$, $R(f; \mathcal{P}, s) = R(f; \mathcal{Q}, t)$.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/pEbD2vi4d8Lr3Rwh6>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

²That is, any point sampled by t belongs to a unique interval $Q \in \mathcal{Q}$.

³That is, any point sampled by t is a cut point of $\mathcal{Q}$.

⁴That is, any point sampled by t is an endpoint of some interval $Q \in \mathcal{Q}$.

[https://www.gradescope.com/courses/1212058/assignments/7390768,](https://www.gradescope.com/courses/1212058/assignments/7390768)

see the Gradescope [documentation](#) for instructions.

5 When to Submit

~~This problem set is due on March 6, 2026 at 11:59 PM.~~

This problem set is due on March 8, 2026 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.